

Text Data in Business and Economics
University of Basel, 21–25 September 2026
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12. Ethical Considerations

Now, we move from “What is?” to “What ought to be?”

- ▶ Most of the content discussed so far concerns positive questions
 - ▶ How does a given model work
 - ▶ What are we measuring, etc.
- ▶ Today, we will discuss the ethics of text-, image- and audio-data related questions
 - ▶ How should models be built from a social viewpoint
 - ▶ How should we (not) use them, etc.
 - ▶ How should textual, visual or audio data (not) represent our society
- ▶ Ethics discussions are necessarily influenced by moral considerations
 - ▶ What is good or bad? Just or unjust

Scientists do not operate outside of society

- ▶ Positive science is interested in discovery and facts (the “truth” about the world)
- ▶ However, positive science interacts with normative considerations in many ways, for example:
 - ▶ Which questions do we deem important to research
 - ▶ What impact will our discovery have on society
 - ▶ What cost are we willing to carry for discovery (not just monetary)
 - ▶ What responsibilities do we (not) have as scientists
 - ▶ Further examples
- ▶ Values and other context-specific characteristics interact with the entire social science pipeline:
 - ▶ Research question → (data collection) → theoretical or empirical analysis → interpretation

Let us discuss a historic case study: Census data

- ▶ A government wants to establish a comprehensive population registration system for administrative and statistical purposes
- ▶ The system should simplify administration and offer perspectives for social research
- ▶ Questions:
 - ▶ Do you have any ethical concerns
 - ▶ Would you be more/less concerned if it is a company/researchers/another actor constructing the comprehensive database? Why?

History has seen horrible examples of unanticipated secondary use

- ▶ During the Second World War: Dutch census data used to facilitate the genocide against Jews and others:
 - ▶ Data collection during peaceful times
 - ▶ Trusted by citizens
 - ▶ Later misuse by the Nazis
 - ▶ Extremely high death rate of Dutch Jews: 73%
- ▶ More: Seltzer, W., & Anderson, M. (2001). The Dark Side of Numbers: The Role of Population Data Systems in Human Rights Abuses. Social Research, 481-513

Ethical Challenges of Generative AI (I/II)

- ▶ **Authorship Attribution, Intellectual Property, Copyright, Authenticity, Academic Integrity**

Plagiarism risks, unclear authorship, ownership of AI-generated content, copyright infringement, and disclosure of AI involvement.

- ▶ **Privacy, Trust, Safety**

Data protection, informed consent, privacy breaches, prompt injection risks.

- ▶ **Bias**

Amplification of social and demographic biases.

- ▶ **Misinformation and Deepfakes**

Generation of realistic false content that undermines trust, democratic discourse, and personal reputation.

Source: Al-kfairy et al. (2024), *Ethical Challenges and Solutions of Generative AI: An Interdisciplinary Perspective*, Informatics.

Ethical Challenges of Generative AI (II/II)

- ▶ **Transparency and Accountability**

Algorithmic opacity, lack of explainability, unclear responsibility for AI-driven harms.

- ▶ **Educational Ethics**

Overreliance on AI, cheating, weakened critical thinking, and inequities in access to AI tools.

- ▶ **Social and Economic Impact**

Job displacement, inequality, concentration of power, and broader societal disruption.

Source: Al-kfairy et al. (2024), *Ethical Challenges and Solutions of Generative AI: An Interdisciplinary Perspective*, Informatics.

Let us discuss further case studies

- ▶ Surveillance technology
 - ▶ Should governments use face recognition technology? If so, for which applications?
 - ▶ How does this question relate to economics research?
- ▶ Representation of different groups (e.g., by ethnicity, gender, age)
 - ▶ From a normative viewpoint, what is the “right” amount and kind of representation of different societal groups in the media
 - ▶ How does this question relate to economics research?
- ▶ Accents and labor market outcomes
 - ▶ (Under what circumstances) should employers be allowed to consider a worker's accent in their hiring decision
 - ▶ How does this question relate to economics research?

How Much Energy Do AI Queries Use?

- ▶ Generative AI requires substantial computing resources.
- ▶ Recent estimates suggest:

Task	Approximate Energy Use
Google search	~0.03–0.3 Wh
ChatGPT prompt	~0.3 Wh
Generating a long answer	~1–3 Wh
Training a frontier LLM	tens of GWh

- ▶ Energy depends strongly on:
 - ▶ model size
 - ▶ number of generated tokens
 - ▶ hardware efficiency
 - ▶ batching of requests

Environmental Impact of Large Language Models

- ▶ LLMs require significant energy at several stages:
 - ▶ **Training** Training frontier models can require tens of gigawatt-hours of electricity.
 - ▶ **Inference** Each prompt requires GPU computation and large data center infrastructure.
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 - ▶ Google's greenhouse gas emissions increased by roughly **50% since 2019**, largely due to AI and data center growth.
- ▶ Environmental impacts include:
 - ▶ electricity demand
 - ▶ carbon emissions
 - ▶ water consumption for cooling
 - ▶ hardware manufacturing and e-waste

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- ▶ **Policy**
 - ▶ Increase transparency in AI energy use
 - ▶ Develop standards for reporting AI carbon footprints